



At Caliola, we built OverKey® to radically simplify the deployment of end-to-end encryption in challenging environmental conditions. OverKey is a mesh virtual private network (VPN) that replaces the traditional hub-and-spoke VPN architecture with a zero-trust, distributed mesh that enables secure and collaborative communications over untrusted and unreliable networks. We have deployed OverKey on top of mesh radios on top of Silvus, TrellisWare, and goTenna; on top of 5G cellular networks; and on top of Starlink.

OverKey – A Mesh VPN for the Department of Defense

Most VPNs adopt a hub-and-spoke or client-server model, wherein secure tunnels are created between each client (spoke) and a centralized server (hub). This allows the client to access resources that are local to the server's network, or to use the server to conceal their identity when accessing external networks. VPNs allow clients to obfuscate their location and evade service provider tracking and eavesdropping.

With a standard hub-and-spoke VPN, all secure communications are mediated by a server. Data that is transmitted from one client to another must be decrypted and re-encrypted at the server along the way. Since there is no end-to-end encryption, VPN servers are central points of failure.

Caliola's OverKey solution replaces this traditional hub-and-spoke architecture with a zero-trust, distributed mesh that enables secure communications over untrusted and unreliable networks. With OverKey, encrypted unicast and multicast traffic flows directly from peer to peer without a server facilitating the connection, and without a server decrypting and re-encrypting the data.

Devices in an OverKey network establish and maintain a secure overlay on top of one or more underlay connectivity services. Caliola has deployed OverKey on top of mesh networks from goTenna, Silvus, and TrellisWare; on top of Starlink; on top of 5G cellular; and on top of the commercial fiber Internet. The overlay presents a flattened and converged common network architecture to users that serves several functions:

1. Nodes must continuously authenticate themselves to each other to maintain membership in the overlay. This provides a zero-trust architecture foundation for OverKey.
2. The overlay distributes the state information required for network address translation. This allows direct peer-to-peer connectivity across wide area networks (WAN).
3. OverKey is unique in its support of multicast (one-to-many) and incast (many-to-one) traffic. The overlay facilitates group communication, even over networks that do not natively support IP multicasting, or WANs that otherwise block IP multicast traffic.
4. The group keying architecture that OverKey uses to support multicast also allows groups of users to be cryptographically segregated into different software-defined virtual local area networks (VLAN) at different security levels – e.g., US-only versus plus-NATO.

5. The overlay facilitates quality-of-service driven routing and traffic prioritization. This allows OverKey to offer Software-Defined WAN (SD-WAN) functionality.

Competing mesh VPN products such as Tailscale and Zero Tier are designed to operate on stable networks and require a consistent connection to a central device for authentication. OverKey was designed specifically to operate at the tactical edge with no reachback connectivity. Caliola has already proven that OverKey can run across networks that are bandwidth limited – e.g., over goTenna radios that transmit just 1200 bits per second – and intermittently connected. OverKey has also proven effective on top of underlay connectivity services that employ vastly different protocol stacks. These features are key differentiators for DoD applications.

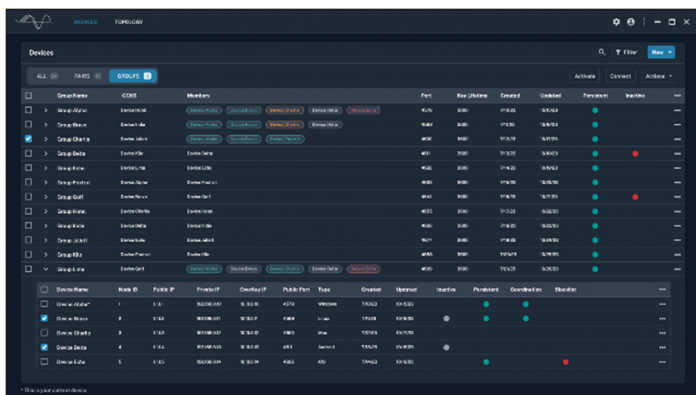
NSA-Approved Protocols and Algorithms

Competing mesh VPN products are built on top of the WireGuard protocol stack, which was developed to circumvent US Government influence on industry standards. OverKey is instead built on top of the IPsec protocol stack and only uses NSA-approved protocols and algorithms.

Authentication and key exchange are performed using the Internet Key Exchange (RFC 2409) and Group Domain of Interpretation (RFCs 6407 and 8263). The Encapsulated Security Payload (RFC 4303) provides data origin authentication, connectionless integrity, an anti-replay service, and traffic flow confidentiality. OverKey only implements the Commercial National Security Algorithm Suite (CNSA) algorithms – AES-256 in Galois Counter Mode; ECDH and ECDSA over Curve P-384; SHA-384; and DH and RSA with a 3072-bit modulus.

As new protocols and algorithms are approved for use in US Government networks, OverKey will be updated accordingly. For example, the IETF Network Working Group is currently developing an extension of RFC 7296 to support group key management. Additionally, Caliola is currently transitioning OverKey to support the CRYSTALS-Kyber and CRYSTALS-Dilithium algorithms for key establishment and digital signatures.

OverKey Management and Control



OverKey networks are managed by a web-based management and control (M&C) interface that exposes the overlay topology and software-defined VLAN structure. Through this interface, an operator can add or delete secure point-to-point (unicast) and group (multicast or incast) connections; force a key refresh; or force the removal of an untrusted node from the overlay via blocklisting. When a node is blocklisted any groups to which it belonged are immediately rekeyed.

OverKey M&C Interface

OverKey as Infrastructure

Finally, Caliola is building network services that leverage the flattened and converged secure network architecture that is presented by OverKey to the Application Layer. For example, our OverKey-powered EdgeSync application enables secure and bandwidth-efficient file synchronization between tactical operating centers and forward-deployed operators. EdgeSync uses OverKey to intelligently and securely route traffic across heterogeneous networks – e.g., so that data is only transmitted once across any bandwidth-constrained bottleneck links. OverKey could also form the foundation of a Data Distribution Service (DDS) like publish-subscribe architecture for use at the tactical edge.